

AI ASSISTED CODING

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BATCH – 04

20 – 01 – 2026

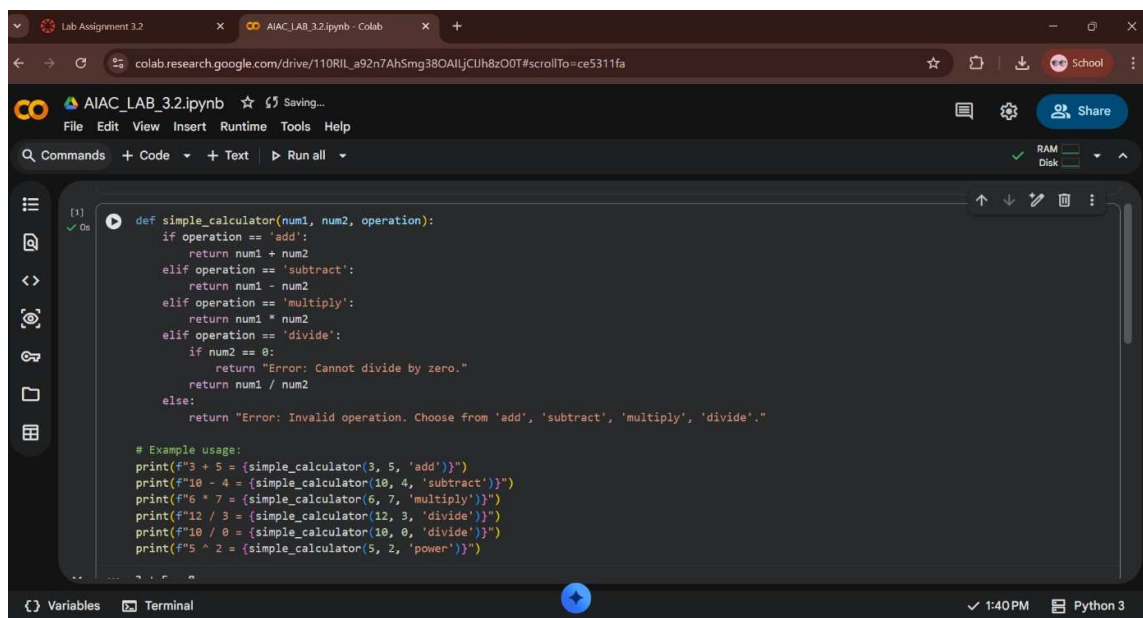
ASSIGNMENT – 3.2

LAB – 03 : Prompt Engineering – Improving Prompts and Context Management.

TASK - 01 : Progressive Prompting for Calculator Design.

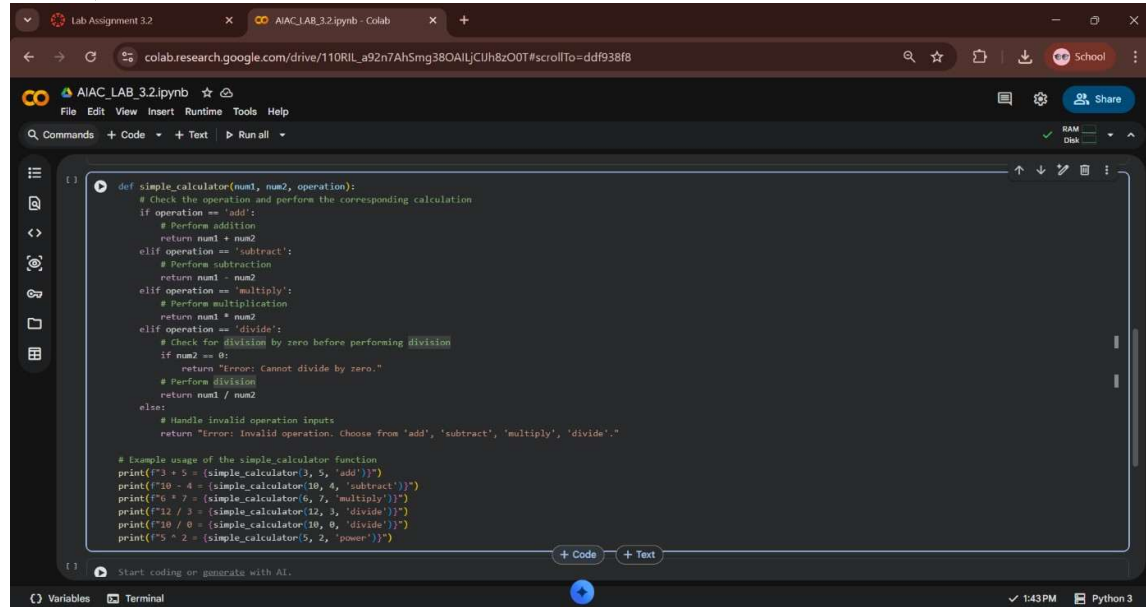
Prompt – 01: Create a Python function named simple_calculator.

Code :

A screenshot of a Google Colab notebook interface. The browser address bar shows a URL from colab.research.google.com. The notebook is titled 'AIAC_LAB_3.2.ipynb' and is in 'Saving...' state. The code editor shows a Python function 'simple_calculator' that takes two numbers and an operation as input. It uses if-elif-else statements to perform addition, subtraction, multiplication, and division. It includes error handling for division by zero and invalid operations. Below the function, there is an example usage section with several print statements demonstrating the function's behavior for different inputs and operations, including a power operation. The bottom of the interface shows tabs for 'Variables' and 'Terminal', and a status bar indicating '1:40 PM' and 'Python 3'.

Prompt – 02 : Create a Python function named simple_calculator that performs basic arithmetic operations (addition, subtraction, multiplication, division). Add comments explaining each step.

Code:

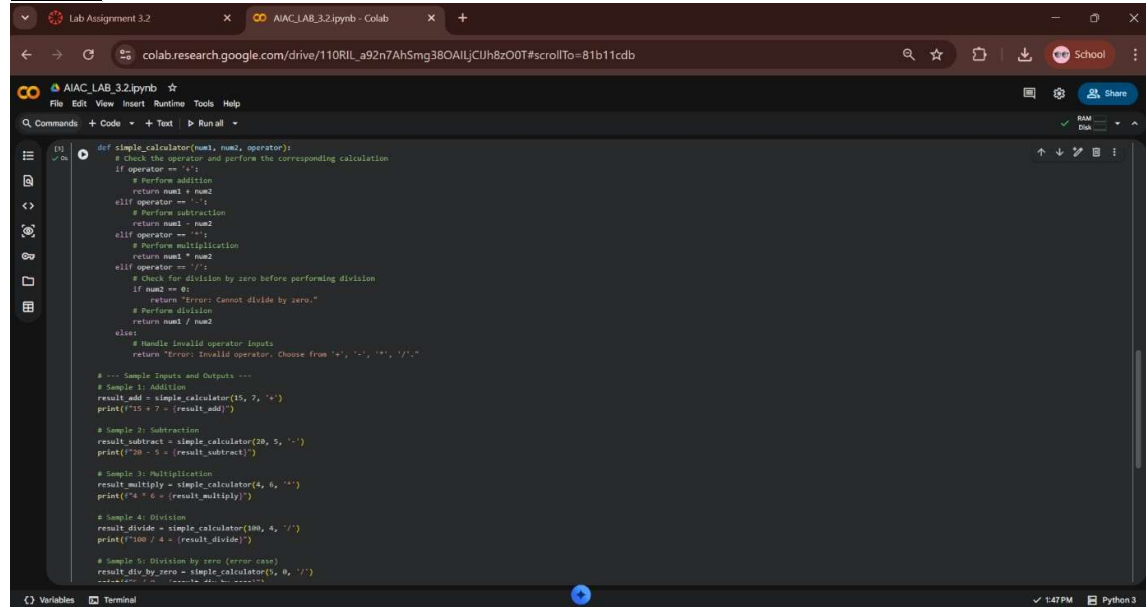


```
def simple_calculator(num1, num2, operation):
    # Check the operation and perform the corresponding calculation
    if operation == 'add':
        # Perform addition
        return num1 + num2
    elif operation == 'subtract':
        # Perform subtraction
        return num1 - num2
    elif operation == 'multiply':
        # Perform multiplication
        return num1 * num2
    elif operation == 'divide':
        # Check for division by zero before performing division
        if num2 == 0:
            return "Error: Cannot divide by zero."
        # Perform division
        return num1 / num2
    else:
        # Handle invalid operation inputs
        return "Error: Invalid operation. Choose from 'add', 'subtract', 'multiply', 'divide'."

# Example usage of the simple_calculator function
print("3 + 5 = (simple_calculator(3, 5, 'add'))")
print("10 - 4 = (simple_calculator(10, 4, 'subtract'))")
print("6 * 7 = (simple_calculator(6, 7, 'multiply'))")
print("12 / 3 = (simple_calculator(12, 3, 'divide'))")
print("10 / 0 = (simple_calculator(10, 0, 'divide'))")
print("5 ^ 2 = (simple_calculator(5, 2, 'power'))")
```

Prompt – 03 : Create a Python function named `simple_calculator` that takes two numbers and an operator (+, -, , /) as input and returns the result. Include comments and provide at least two sample inputs with outputs.

Code :



```
def simple_calculator(num1, num2, operator):
    # Check the operator and perform the corresponding calculation
    if operator == '+':
        # Perform addition
        return num1 + num2
    elif operator == '-':
        # Perform subtraction
        return num1 - num2
    elif operator == '*':
        # Perform multiplication
        return num1 * num2
    elif operator == '/':
        # Check for division by zero before performing division
        if num2 == 0:
            return "Error: Cannot divide by zero."
        # Perform division
        return num1 / num2
    else:
        # Handle invalid operator inputs
        return "Error: Invalid operator. Choose from '+', '-', '*', '/'."

# --- Sample Inputs and Outputs ---
# Sample 1: Addition
result_add = simple_calculator(15, 7, '+')
print("15 + 7 = (result_add)")

# Sample 2: Subtraction
result_subtract = simple_calculator(20, 5, '-')
print("20 - 5 = (result_subtract)")

# Sample 3: Multiplication
result_multiply = simple_calculator(4, 6, '*')
print("4 * 6 = (result_multiply)")

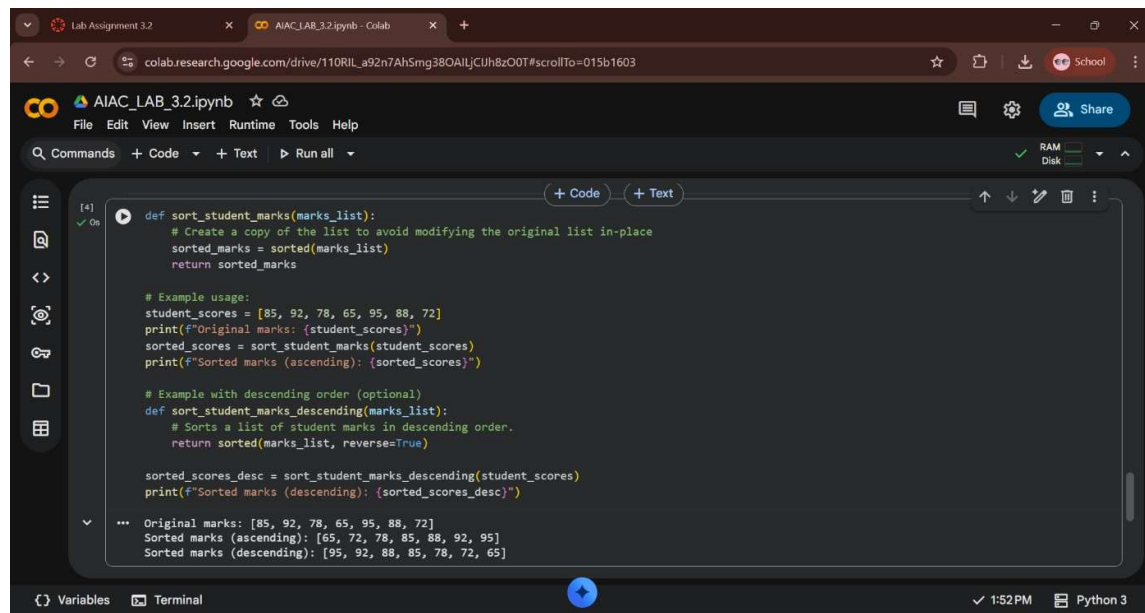
# Sample 4: Division
result_divide = simple_calculator(100, 4, '/')
print("100 / 4 = (result_divide)")

# Sample 5: Division by zero (error case)
result_div_by_zero = simple_calculator(5, 0, '/')
print("5 / 0 = (result_div_by_zero) Error: Cannot divide by zero.")
```

TASK – 02 : Refining Prompts for Sorting Logic

Prompt – 01 : Write a Python Function to sort the Student Marks.

Code :

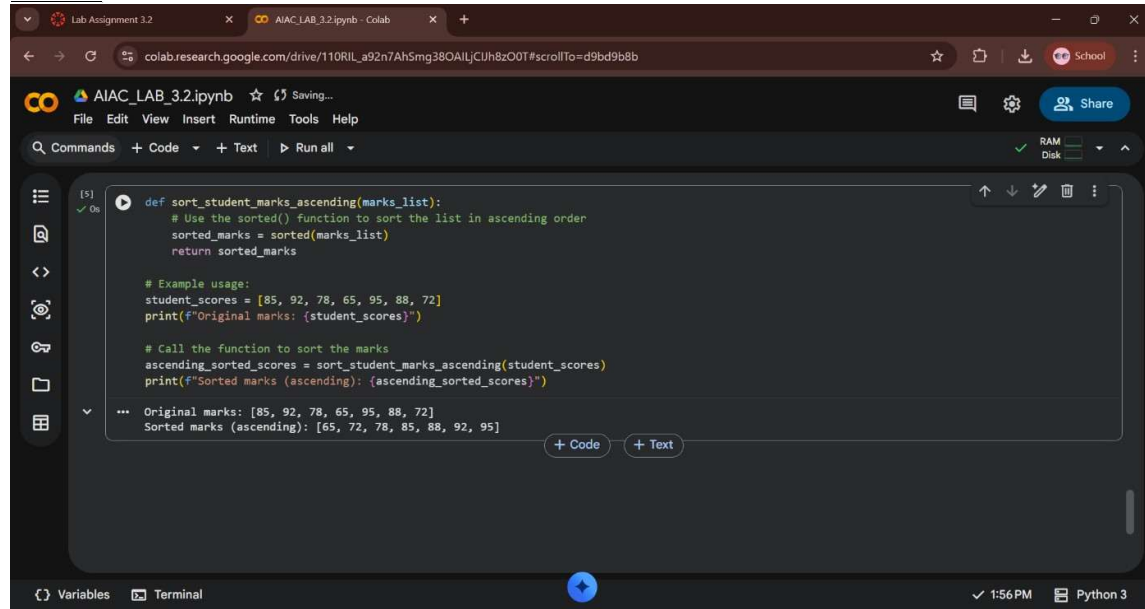


The screenshot shows a Google Colab notebook titled "AIAC_LAB_3.2.ipynb". The code defines a function `sort_student_marks(marks_list)` that creates a copy of the list and returns it sorted in ascending order. It also includes an example usage and a function for descending order sorting.

```
[4] def sort_student_marks(marks_list):  
    # Create a copy of the list to avoid modifying the original list in-place  
    sorted_marks = sorted(marks_list)  
    return sorted_marks  
  
# Example usage:  
student_scores = [85, 92, 78, 65, 95, 88, 72]  
print(f"Original marks: {student_scores}")  
sorted_scores = sort_student_marks(student_scores)  
print(f"Sorted marks (ascending): {sorted_scores}")  
  
# Example with descending order (optional)  
def sort_student_marks_descending(marks_list):  
    # Sorts a list of student marks in descending order.  
    return sorted(marks_list, reverse=True)  
  
sorted_scores_desc = sort_student_marks_descending(student_scores)  
print(f"Sorted marks (descending): {sorted_scores_desc}")  
  
*** Original marks: [85, 92, 78, 65, 95, 88, 72]  
Sorted marks (ascending): [65, 72, 78, 85, 88, 92, 95]  
Sorted marks (descending): [95, 92, 88, 85, 78, 72, 65]
```

Prompt – 02 : Write a Python function that takes a list of student marks and sorts them in ascending order.

Code :

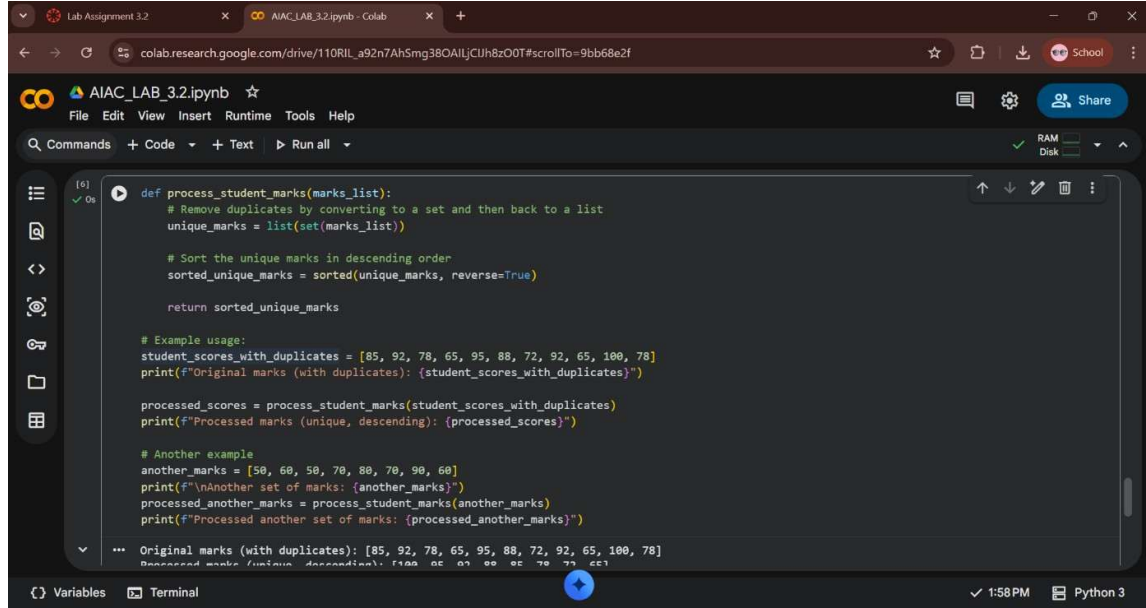


The screenshot shows a Google Colab notebook titled "AIAC_LAB_3.2.ipynb". The code defines a function `sort_student_marks_ascending(marks_list)` that uses the `sorted()` function to sort the list in ascending order. It includes an example usage and a call to the function.

```
[5] def sort_student_marks_ascending(marks_list):  
    # Use the sorted() function to sort the list in ascending order  
    sorted_marks = sorted(marks_list)  
    return sorted_marks  
  
# Example usage:  
student_scores = [85, 92, 78, 65, 95, 88, 72]  
print(f"Original marks: {student_scores}")  
  
# Call the function to sort the marks  
ascending_sorted_scores = sort_student_marks_ascending(student_scores)  
print(f"Sorted marks (ascending): {ascending_sorted_scores}")  
  
*** Original marks: [85, 92, 78, 65, 95, 88, 72]  
Sorted marks (ascending): [65, 72, 78, 85, 88, 92, 95]
```

Prompt – 03 : Write an efficient Python function that takes a list of student marks (0–100), sorts them in descending order, removes duplicates, and returns the sorted list.

Code :

A screenshot of a Google Colab notebook interface. The browser address bar shows a Google Drive link. The notebook is titled 'AIAC_LAB_3.2.ipynb'. The code editor contains a Python function 'process_student_marks' that takes a list of marks, removes duplicates by converting to a set and back to a list, sorts in descending order, and returns the sorted list. Below the function, there are two example usages with print statements. The first example uses a list with duplicates, and the second uses a list with unique marks. The output of the first example is visible at the bottom of the code cell, showing the original and processed marks.

```
[6] ✓ Os def process_student_marks(marks_list):  
    # Remove duplicates by converting to a set and then back to a list  
    unique_marks = list(set(marks_list))  
  
    # Sort the unique marks in descending order  
    sorted_unique_marks = sorted(unique_marks, reverse=True)  
  
    return sorted_unique_marks  
  
# Example usage:  
student_scores_with_duplicates = [85, 92, 78, 65, 95, 88, 72, 92, 65, 100, 78]  
print(f"Original marks (with duplicates): {student_scores_with_duplicates}")  
  
processed_scores = process_student_marks(student_scores_with_duplicates)  
print(f"Processed marks (unique, descending): {processed_scores}")  
  
# Another example  
another_marks = [50, 60, 50, 70, 80, 70, 90, 60]  
print(f"Another set of marks: {another_marks}")  
processed_another_marks = process_student_marks(another_marks)  
print(f"Processed another set of marks: {processed_another_marks}")  
  
... Original marks (with duplicates): [85, 92, 78, 65, 95, 88, 72, 92, 65, 100, 78]  
Processed marks (unique, descending): [100, 95, 92, 88, 85, 78, 72, 65]
```

TASK – 03 : Few-Shot Prompting for Prime Number Validation. **Prompt :** Write a Python function is_prime(n) that checks whether a given is Prime or Not, Using Examples:

I/P: 2 -> O/P True

I/P: 4 -> O/P False

Handle edge cases like negative numbers and 0 properly.

Code:

```
def is_prime(n):  
    # Handle edge cases: numbers less than or equal to 1 are not prime  
    if n <= 1:  
        return False  
    # 2 is the only even prime number  
    if n == 2:  
        return True  
    # All other even numbers are not prime  
    if n % 2 == 0:  
        return False  
    # Check for divisibility from 3 up to the square root of n  
    # Only check odd numbers  
    i = 3  
    while i * i <= n:  
        if n % i == 0:  
            return False  
        i += 2  
    return True  
  
# Example usage:  
print(f"Is 2 prime? (is_prime(2))") # Expected: True  
print(f"Is 4 prime? (is_prime(4))") # Expected: False  
print(f"Is 1 prime? (is_prime(1))") # Expected: False  
print(f"Is 17 prime? (is_prime(17))") # Expected: True  
print(f"Is 0 prime? (is_prime(0))") # Expected: False  
print(f"Is -5 prime? (is_prime(-5))") # Expected: False  
print(f"Is 29 prime? (is_prime(29))") # Expected: True  
print(f"Is 33 prime? (is_prime(33))") # Expected: False
```

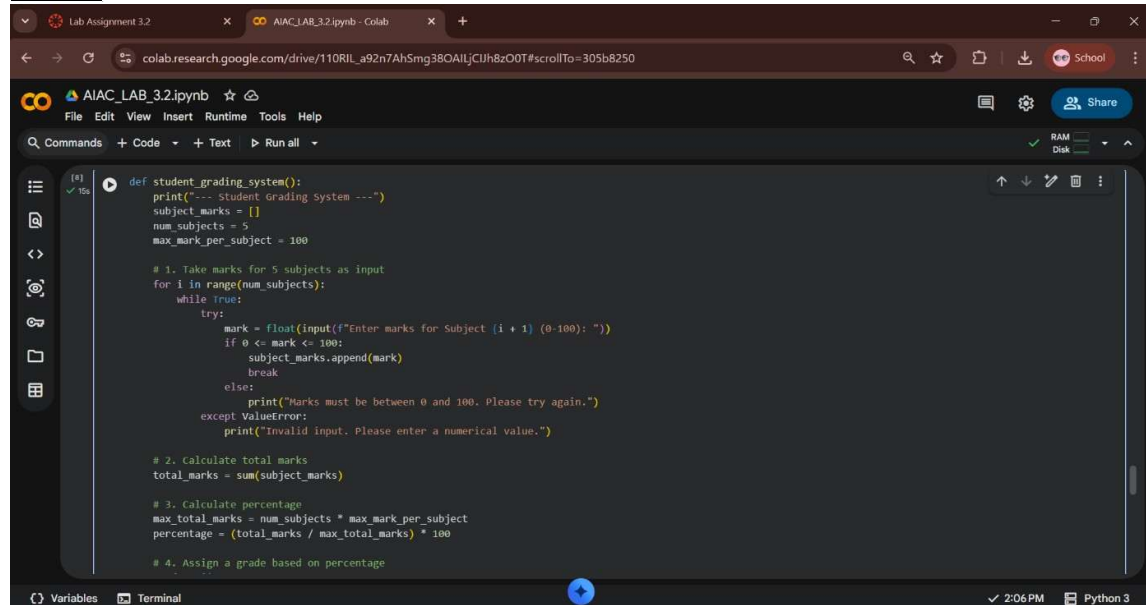
TASK – 04 : Prompt-Guided UI Design for Student Grading System.

Prompt : Design a simple Python-based user interface (CLI or GUI) for a student grading system.

The program should:

- Take marks for 5 subjects as input
- Calculate total marks
- Calculate percentage
- Assign a grade based on percentage
- Display all results clearly.

Code:



The screenshot shows a Google Colab notebook titled "AIAC_LAB_3.2.ipynb". The code defines a function `student_grading_system()` that takes marks for 5 subjects as input, calculates the total marks, percentage, and assigns a grade based on the percentage. The code is as follows:

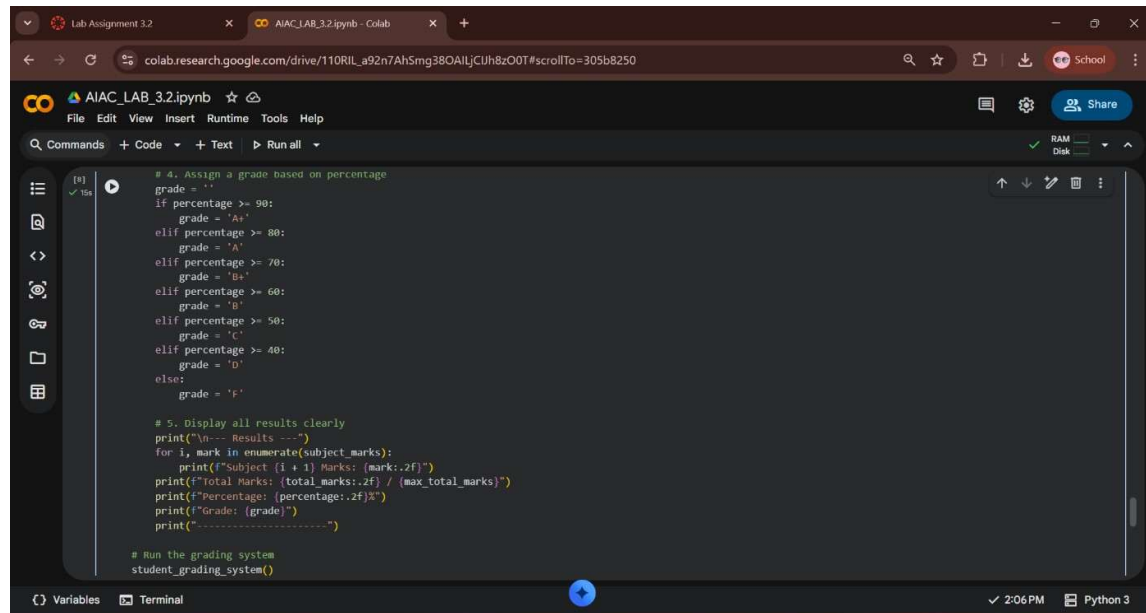
```
[9] def student_grading_system():
    print("--- Student Grading System ---")
    subject_marks = []
    num_subjects = 5
    max_mark_per_subject = 100

    # 1. Take marks for 5 subjects as input
    for i in range(num_subjects):
        while True:
            try:
                mark = float(input(f"Enter marks for Subject {i + 1} (0-100): "))
                if 0 <= mark <= 100:
                    subject_marks.append(mark)
                    break
            except ValueError:
                print("Marks must be between 0 and 100. Please try again.")
            print("Invalid input. Please enter a numerical value.")

    # 2. Calculate total marks
    total_marks = sum(subject_marks)

    # 3. Calculate percentage
    max_total_marks = num_subjects * max_mark_per_subject
    percentage = (total_marks / max_total_marks) * 100

    # 4. Assign a grade based on percentage
```



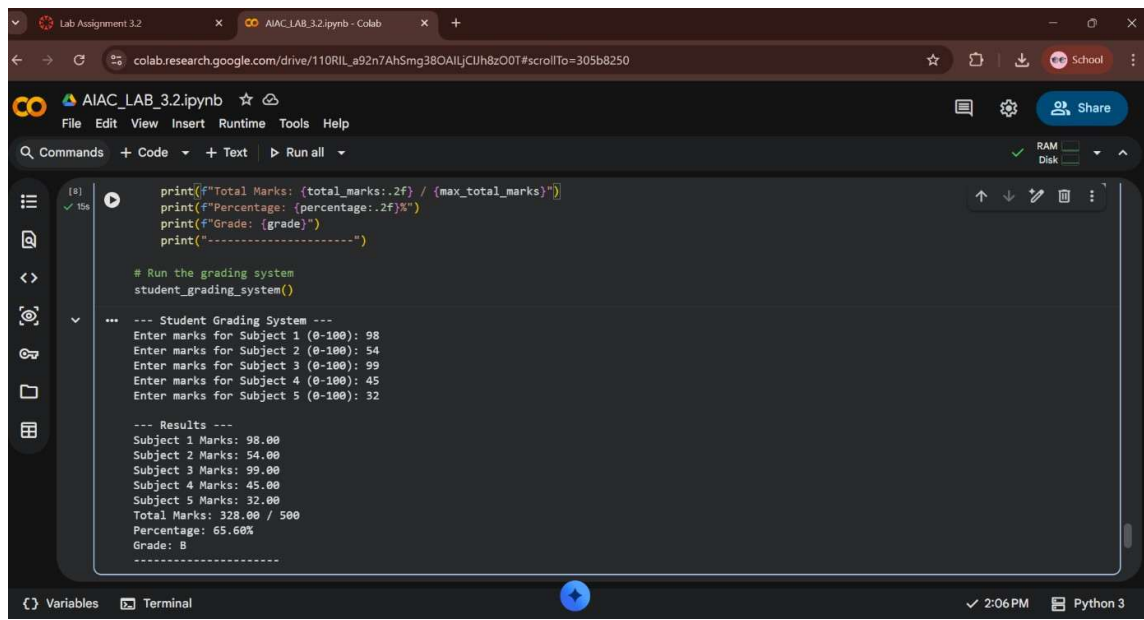
The screenshot shows the continuation of the Python script in the Google Colab notebook. The code defines the grade assignment logic based on the percentage and displays the results for each subject, total marks, percentage, and grade. The code is as follows:

```
[9] # 4. Assign a grade based on percentage
grade = ''
if percentage >= 90:
    grade = 'A+'
elif percentage >= 80:
    grade = 'A'
elif percentage >= 70:
    grade = 'B+'
elif percentage >= 60:
    grade = 'B'
elif percentage >= 50:
    grade = 'C'
elif percentage >= 40:
    grade = 'D'
else:
    grade = 'F'

# 5. Display all results clearly
print("\n--- Results ---")
for i, mark in enumerate(subject_marks):
    print(f"Subject {i + 1} Marks: {mark:.2f}")
print(f"Total Marks: {total_marks:.2f} / {max_total_marks}")
print(f"Percentage: {percentage:.2f}%")
print(f"Grade: {grade}")
print("-----")

# Run the grading system
student_grading_system()
```

Output:

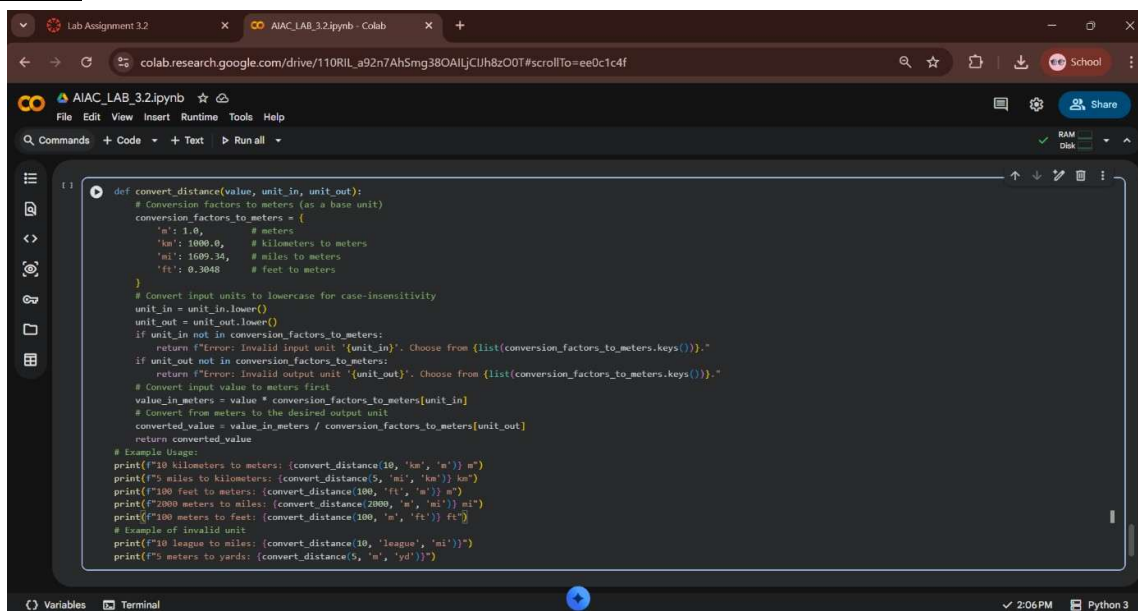


```
[8] ✓ 15s  
print(f"Total Marks: {total_marks:.2f} / {max_total_marks}")  
print(f"Percentage: {percentage:.2f}%")  
print(f"Grade: {grade}")  
print("-----")  
  
# Run the grading system  
student_grading_system()  
  
--- Student Grading System ---  
Enter marks for Subject 1 (0-100): 98  
Enter marks for Subject 2 (0-100): 54  
Enter marks for Subject 3 (0-100): 99  
Enter marks for Subject 4 (0-100): 45  
Enter marks for Subject 5 (0-100): 32  
  
--- Results ---  
Subject 1 Marks: 98.00  
Subject 2 Marks: 54.00  
Subject 3 Marks: 99.00  
Subject 4 Marks: 45.00  
Subject 5 Marks: 32.00  
Total Marks: 328.00 / 500  
Percentage: 65.60%  
Grade: B  
-----
```

TASK – 05 : Analyzing Prompt Specificity in Unit Conversion Functions.

Prompt – 01 : Write a function to convert distance.

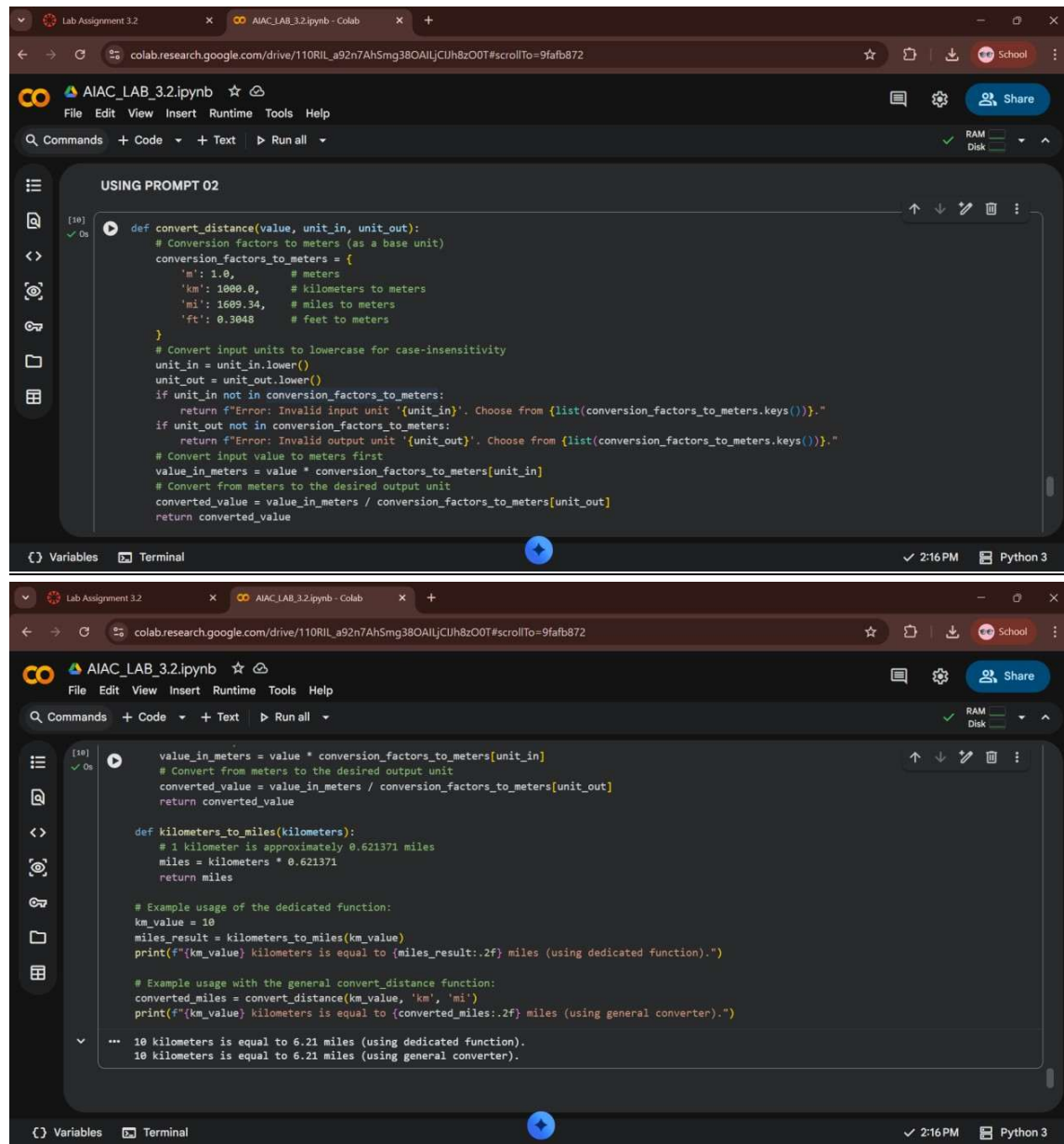
Code:



```
1.1 def convert_distance(value, unit_in, unit_out):  
    # Conversion factors to meters (as a base unit)  
    conversion_factors_to_meters = {  
        'm': 1.0,           # meters  
        'km': 1000.0,       # kilometers to meters  
        'mi': 1609.34,      # miles to meters  
        'ft': 0.3048        # feet to meters  
    }  
    # Convert input units to lowercase for case-insensitivity  
    unit_in = unit_in.lower()  
    unit_out = unit_out.lower()  
    if unit_in not in conversion_factors_to_meters:  
        return f"Error: Invalid input unit '{unit_in}'. Choose from {list(conversion_factors_to_meters.keys())}."  
    if unit_out not in conversion_factors_to_meters:  
        return f"Error: Invalid output unit '{unit_out}'. Choose from {list(conversion_factors_to_meters.keys())}."  
    # Convert input value to meters first  
    value_in_meters = value * conversion_factors_to_meters[unit_in]  
    # Convert from meters to the desired output unit  
    converted_value = value_in_meters / conversion_factors_to_meters[unit_out]  
    return converted_value  
  
# Example Usage:  
print(f"10 kilometers to meters: {convert_distance(10, 'km', 'm')} m")  
print(f"5 miles to kilometers: {convert_distance(5, 'mi', 'km')} km")  
print(f"100 feet to meters: {convert_distance(100, 'ft', 'm')} m")  
print(f"2000 meters to miles: {convert_distance(2000, 'm', 'mi')} mi")  
print(f"100 meters to feet: {convert_distance(100, 'm', 'ft')} ft")  
# Example of invalid unit  
print(f"10 league to miles: {convert_distance(10, 'league', 'mi')}")  
print(f"5 meters to yards: {convert_distance(5, 'm', 'yd')}")
```


Prompt – 02: Write a Python function to convert kilometers to miles.

Code:



The image displays two screenshots of a Google Colab notebook titled 'AIAC_LAB_3.2.ipynb'. The first screenshot shows the definition of a general function `convert_distance` that takes a value, an input unit, and an output unit. It uses a dictionary of conversion factors to meters as a base unit. The second screenshot shows the execution of the code, which includes a dedicated function `kilometers_to_miles` and example usage for both functions, resulting in printed output.

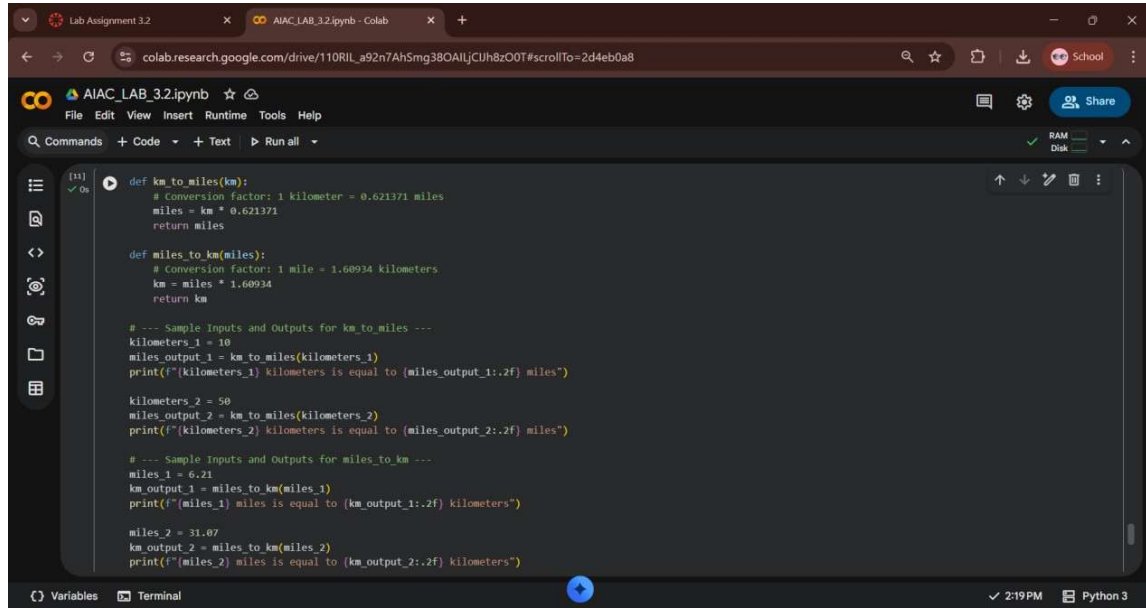
```
[10]: def convert_distance(value, unit_in, unit_out):  
      # Conversion factors to meters (as a base unit)  
      conversion_factors_to_meters = {  
          'm': 1.0,           # meters  
          'km': 1000.0,       # kilometers to meters  
          'mi': 1609.34,      # miles to meters  
          'ft': 0.3048        # feet to meters  
      }  
      # Convert input units to lowercase for case-insensitivity  
      unit_in = unit_in.lower()  
      unit_out = unit_out.lower()  
      if unit_in not in conversion_factors_to_meters:  
          return f"Error: Invalid input unit '{unit_in}'. Choose from {list(conversion_factors_to_meters.keys())}."  
      if unit_out not in conversion_factors_to_meters:  
          return f"Error: Invalid output unit '{unit_out}'. Choose from {list(conversion_factors_to_meters.keys())}."  
      # Convert input value to meters first  
      value_in_meters = value * conversion_factors_to_meters[unit_in]  
      # Convert from meters to the desired output unit  
      converted_value = value_in_meters / conversion_factors_to_meters[unit_out]  
      return converted_value  
  
      value_in_meters = value * conversion_factors_to_meters[unit_in]  
      # Convert from meters to the desired output unit  
      converted_value = value_in_meters / conversion_factors_to_meters[unit_out]  
      return converted_value  
  
      def kilometers_to_miles(kilometers):  
          # 1 kilometer is approximately 0.621371 miles  
          miles = kilometers * 0.621371  
          return miles  
  
      # Example usage of the dedicated function:  
      km_value = 10  
      miles_result = kilometers_to_miles(km_value)  
      print(f"{km_value} kilometers is equal to {miles_result:.2f} miles (using dedicated function).")  
  
      # Example usage with the general convert_distance function:  
      converted_miles = convert_distance(km_value, 'km', 'mi')  
      print(f"{km_value} kilometers is equal to {converted_miles:.2f} miles (using general converter).")  
  
      ...  
      10 kilometers is equal to 6.21 miles (using dedicated function).  
      10 kilometers is equal to 6.21 miles (using general converter).
```

Prompt – 03: Write two Python functions:

1. `km_to_miles(km)`
2. `miles_to_km(miles)`

Use correct conversion factors, add comments, and provide sample inputs and outputs.

Code:



```
[11] def km_to_miles(km):  
    # Conversion factor: 1 kilometer = 0.621371 miles  
    miles = km * 0.621371  
    return miles  
  
def miles_to_km(miles):  
    # Conversion factor: 1 mile = 1.60934 kilometers  
    km = miles * 1.60934  
    return km  
  
# --- Sample Inputs and Outputs for km_to_miles ---  
kilometers_1 = 10  
miles_output_1 = km_to_miles(kilometers_1)  
print(f"{kilometers_1} kilometers is equal to {miles_output_1:.2f} miles")  
  
kilometers_2 = 50  
miles_output_2 = km_to_miles(kilometers_2)  
print(f"{kilometers_2} kilometers is equal to {miles_output_2:.2f} miles")  
  
# --- Sample Inputs and Outputs for miles_to_km ---  
miles_1 = 6.21  
km_output_1 = miles_to_km(miles_1)  
print(f"{miles_1} miles is equal to {km_output_1:.2f} kilometers")  
  
miles_2 = 31.07  
km_output_2 = miles_to_km(miles_2)  
print(f"{miles_2} miles is equal to {km_output_2:.2f} kilometers")
```

THANK YOU!!